PREPARATION OF LAND USE/ LAND COVER MAP OF DEHRADUN CITY AND ACCURACY ASSESSMENT USING REMOTE SENSING AND GIS

Dissertation submitted to the

University of Delhi,
In partial fulfilment of the requirements for the
Award of the Degree of

Master of Science

In

Geology

By

Rishant Prakash

Under the Supervision of

Prof. Vimal Singh

DEPARTMENT OF GEOLOGY
CENTRE FOR ADVANCED STUDIES
UNIVERSITY OF DELHI-110007
JUNE-2021



Phone: +91-011-27667073; 27667725 Ext. 1442-1443
Telefax: +91-011-27666295. Web-www.du.ac.in

CERTIFICATE

I, solemnly declare that the research conducted on the topic entitled

"**Preparation of Land Use/Land Cover map of Dehradun City and Accuracy Assessment using Remote Sensing and GIS"** is original and bona fide piece of my dissertation work.

I, further declare that, the results of the research work mentioned in this dissertation have not been submitted in part or full for the award of any other degree or diploma at any university.

Rishant Prakash

Prof. VIMAL SINGH
Supervisor

Department of Geology
Centre of Advanced Studies
University of Delhi
Delhi 110007

Prof. ANUPAM CHATTOPADHYAY
Head of the department

Department of Geology
Centre of Advanced Studies
University of Delhi
Delhi 110007

ACKNOWLEDGMENT

I would like to take this moment to extend my deepest thanks to my guide and mentor **Prof. Vimal Singh** for providing me this great work opportunity and introducing me to the recent advances in the field of Remote Sensing and GIS. Thank you for providing the perfect study environment, resources, and stoic patience that led to the completion of this report. Without his guidance and persistent help, this dissertation would not have been possible.

I am also obliged to the esteemed Head of the Department of Geology**, Prof. Anupam Chattopadhyay** for fostering and maintaining an environment of research in the department. It is my radiant sentiment to place on record my sincere thanks to the **Department of Geology**, University of Delhi, the esteemed organization that allowed me to use my time in a channelized manner towards learning and exploring.

I would like to particularly thank Mr. Vicky Shankar for being a constant guiding light during the course of my dissertation. An appreciable amount of support also goes to Mr. Parv Kasana and Mr. Arkaprabha Sarkar without whom I would not be able to work and gain confidence in my work.

This year has not been a usual one. These few months have tested our patience, strength, and determination to fight a global pandemic. Learning in a home environment is way different from the classrooms or lab. Keeping this in the note, my deepest appreciation goes to my parents and family members for providing me with the necessary resources and support. Their love and guidance are with me in whatever I pursue.

RISHANT PRAKASH

TABLE OF CONTENTS:-

Page No.

Chapter – 1

INTRODUCTION **7-9**

Chapter – 2

GEOLOGICAL AND GEOMORPHOLOGICAL SETTING OF THE STUDY AREA **10-11**

Chapter-3

METHODOLOGY **12-18**

Chapter-4

RESULTS AND DISCUSSION **19-30**

Chapter-5

CONCLUSION **31**

REFERENCES **32-34**

LIST OF FIGURES:

Figure 1: Location map of the study area -Dehradun City

Figure 2: Location map with regional geology and tectonics of Dehradun region modified from Mahajan 2000.

Figure 3 showing: Schematic of workflow of LULC followed by accuracy assessment

Figure 4: Classified map of the study area for the year 2021.

Figure 5: Ground Points on the study area

Figure 6: (a) LULC map of Dehradun City-2001; (b) graph showing distribution of various classes.

Figure 7: (a) LULC map of Dehradun City-2011; (b) graph showing distribution of various classes.

Figure 8: (a) LULC map of Dehradun City-2021; (b) graph showing distribution of various classes.

Figure 9: Graph showing comparison of area covered by various classes in three different years.

Figure 10: Graph showing percentage change of area by different classes between 2001 and 2011.

Figure 11: Graph showing percentage change of area by different classes between 2011 and 2021.

Figure 12: Graph showing percentage change of area by different classes between 2001 and 2021.

LIST OF TABLES

Table 1: Details of Landsat data used for classification

Table 2: Landcover classification scheme

Table 3: Error matrix of the LULC classification of the year 2011

Table 4- Land Use/Cover of 2001

Table 5 – Accuracy Assessment report of LULC 2001.

Table 6- Land Use/Cover of 2011

Table 7 – Accuracy Assessment report of LULC 2011.

Table 8- Land Use/Cover of 2021

Table 9– Accuracy Assessment report of LULC 2021.

Table- 10: Area covered by different classes in various years.

Table 11:- Percentage change in different classes between 2001 and 2011.

Table 12:- Percentage change in different classes between 2011 and 2021.

Table 13:- Percentage change in different classes between 2001 and 2021.

CHAPTER-1

INTRODUCTION

1.1 Objective:-

To Prepare the Land Use / Land Cover map of Dehradun City and Accuracy Assessment using Remote Sensing and GIS.

1.2 Background:-

Remote sensing is a robust tool for the regional mapping of geological features and natural resources. Starting with the utilization of image during the first stages of development of remote sensing within the mid- 1970s, enough progress has been achieved in the data interpretation with the easy availability of digital data. Digital processing of remote sensing data has gained a faster pace in the last 10 years. In India, with the establishment of remote sensing centers all over the country in recent years, attention has been focused on the massive scale processing for natural resources evaluation.

One important aspect in remote sensing is that the characterization and classification of spectral measurements taken from satellites into various features of the land surface. Pattern recognition will get disbursed if appropriate procedures are adopted for classification. Several classification methods have been developed within the field of statistics but their applicability to the processing of knowledge is limited because of the spatial variation of natural features. In classification studies, it is desirable to grasp how well the classes may get separated by observing the values of some feature vector for a collection of samples. In other words, one wants to understand what quantity information the features provide for distinguishing the classes. To answer these questions a measure is required to quantify the quantity of data on the features.

The objective of this study is to produce Land use Land Cover map of Dehradun City using supervised classification with the accuracy assessment. This study utilizes a multiband data set to work out the effectiveness in improving the classification. The process of supervised classification is most widely used for quantitative analyses of remote sensing data. The focus is on the use of supervised classification using **maximum likelihood classifier** and therefore the derivation of meaningful confidence level to any or all land cover classes.

1.3 Study Area:-

Dehradun is a medium-sized class I city nestled within the foothills of the Himalayas (Dipanwita Dutta, 2015). Uttarakhand is divided into 13 districts and two regions. Kumaon and Garhwal are the two regions. Dehradun is a district located in the Garhwal Himalaya. The city is further divided into sub-divisions and blocks. The city has a total size of around 65 square kilometers.

The city lies between 78°00' to 78°10' East longitudes and 30°15' to 30°25' North latitude. It is located at an altitude of 640 m above mean sea level (Jain and Jain, 2006). The northern part is covered by the higher-lesser Himalayan ranges, (elevation > 1000m) and southern part is covered by younger Shivalik hills (elevation reaching up to 700-800m). The city is bounded by dense forest in the south and divides by a large no. of seasonal streams. The Rispana and the Bindal rivers mark the eastern and the western boundary of the city respectively. The city is located in an intermontane valley, locally called as Doon (also spelled as Dun). The temperature ranges varies from 36° C to 16.7° C during summer months and from 23.4° C to 5.2° C during the winter months. The average annual rainfall in the city is 200 cm during monsoon period (i.e. June to August) (Singh et al., 2013). The scenic beauty and calm weather of the city, therefore attracts plenty of tourists.

After being declared as the state capital in the year 2000, the town experienced an outsized inflow of immigrants mainly because of economic reasons and better livelihood. Since the city has offered a lot of job opportunities, it's been a favourite destination for the unemployed and landless people of surrounding districts. So as to accommodate this massive migrant population, the town has undergone a rapid expansion by converting the agricultural and forested land into built-up areas, etc. (Bindu Agarwal, 2018). Therefore, the city Dehradun has been taken as the study area for the preparation of Land Use Land Cover map.

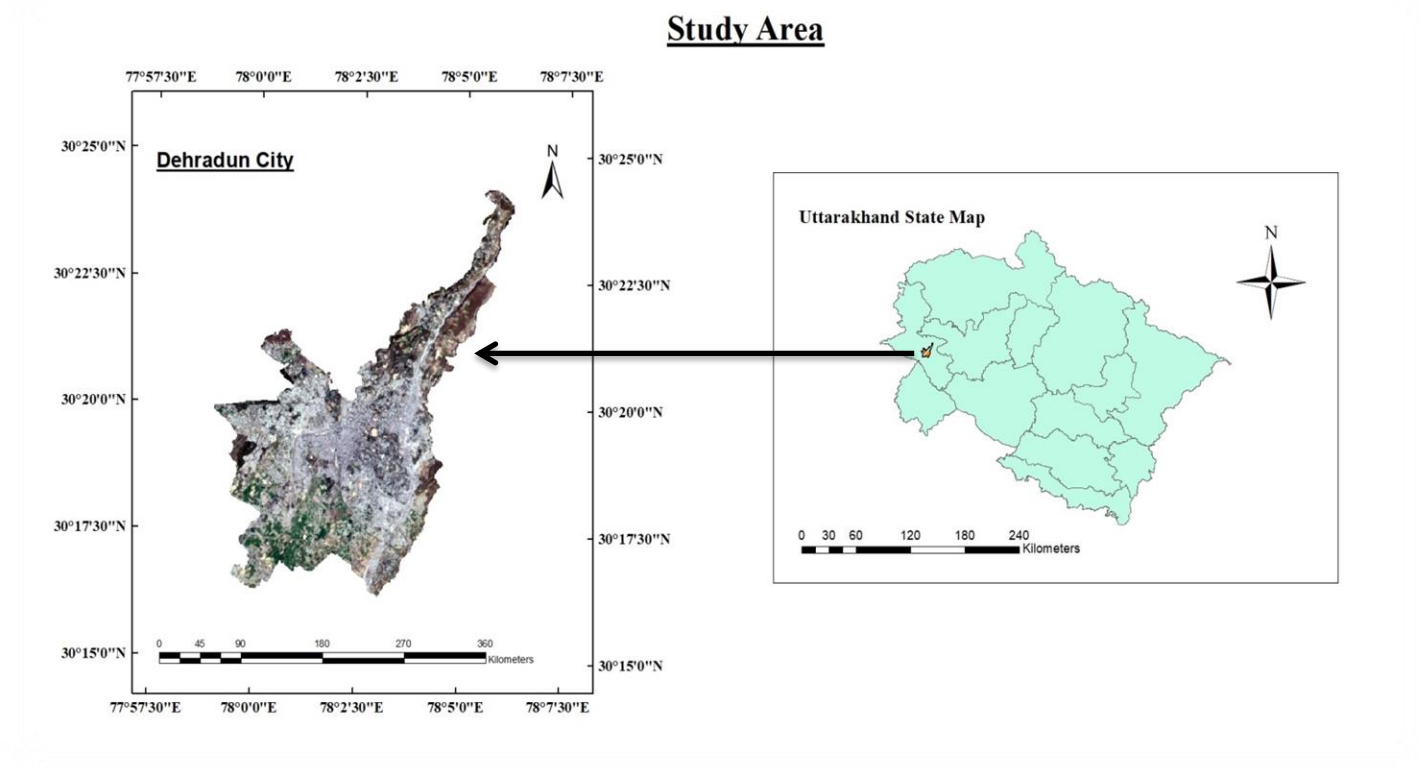


Figure 1: Location map of the study area -Dehradun City

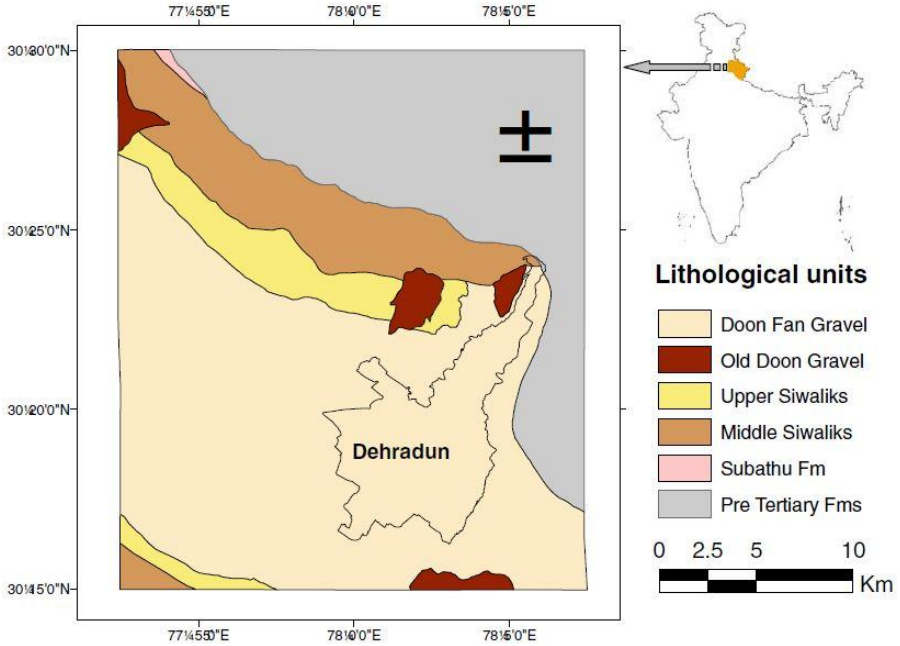
Chapter 2

GEOLOGICAL AND GEOMORPHOLOGICAL SETTING OF THE STUDY AREA

The Dehra Dun is one of the largest intermontane valleys, stretching between the Mussoorie Ranges of the Lesser Himalaya to the north and the Siwalik Ranges to the south (Fig. 2). The Siwalik ranges predominantly comprise of Middle Siwalik (sandstone), and Upper Siwalik (conglomerates) (Swati Sinha, 2016). The Main Boundary Thrust (MBT) to the north, the Himalayan Frontal Thrust (HFT) to the south, the Yamuna Tear Fault (YTF) to the west, and the Ganga Tear Fault (GTF) to the east, all surround the Dehra Dun, making it a structurally isolated block (Fig. 2). The Ganga and Yamuna rivers flow through the Siwalik Ranges, respectively, along the GTF and YTF transverse faults (Karunakaran and Rao, 1979; Raiverman et al., 1994). The intermontane valley is 80 kilometers long and 25 kilometers wide, with a NW-SE trend. With a softer (0–5°) SW slope and a steeper (0–10°) NE slope, it is an asymmetrical valley. The principal geomorphic units in this region include alluvial fans, hillocks, river terraces, and floodplains, with the valley fills referred to as 'Dun gravels' which overlay the Siwalik rocks (Nossin, 1971; Nakata, 1972; Singh et al., 2001; Thakur and Pandey, 2004; Thakur et al., 2007). Doon gravels consist of gravel beds that become coarse with depth, with occasional clay lenses and layers. In the northern part of Dehradun City 'Doon gravels' are absent, where Siwalik rocks are exposed, and increase in thickness towards the south (Fig. no. 2). In the middle of Doon valley the maximum depth of the Doon gravels is estimated to be several hundred meters. However, it resembles the underlying upper Siwalik rocks dominated by boulder conglomerates and sandstones at depths of more than 40-50 m (Auden 1936). Due to possible lime enrichment from the nearby limestone of the Lesser Himalaya (Fig. 2), the Doon gravels are cemented in some locations. This cemented layer varies from 20–25 m in depth in the northern and central part to more than 30 m in the south-southwestern part of the city, whereas in the southeastern side, the depth is lesser than 25 m.

Nossin (1971) documented three large fans made by streams originating from the Mussoorie ranges'. Four fill surfaces were observed by Nakata (1972, 1975), who interpreted them as the result of uneven uplift within the valley.

The main thrust that separates the Shivaliks from the Lesser Himalaya in the north is known as the Main Boundary thrust. The Santaurgarh thrust is another notable structural structure that has been described in this area (Raiverman et al., 1994).



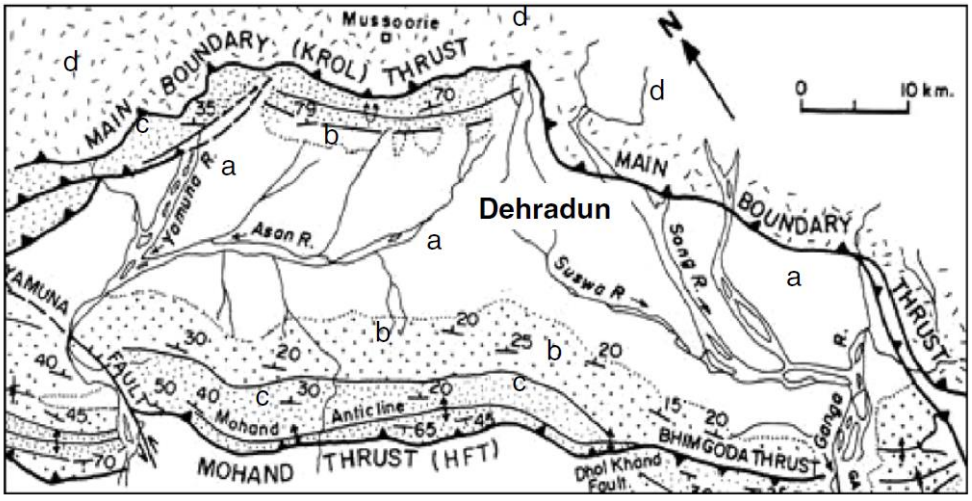


Figure 2: Location map with regional geology and tectonics of Dehradun region modified from Mahajan 2000.

Geomorphologically, the hilltop surface (residual hill) and the piedmont surface are the two major geomorphic surfaces identified in the Dehradun area (Singh et al. 2004). The Piedmont surface can be further divided into middle Doon surface (MDS) and lower Doon surface (LDS) (Nossin 1971 and Nakata 1972). The hilltop surface is made up of thick boulder and gravel beds, with boulders up to 2 meters in diameter, found on the flat crest of the residual hills in the city's northernmost reaches. Both the LDS and MDS piedmont surfaces have less consolidated and weathered gravel beds. The LDS (piedmont terrace), which is largely located in the central part of the Doon valley, is the lowest alluvial fan deposit of the Ganga and Yamuna rivers' major tributaries. The LDS deposits are less than 10 m thick and are underlain by the middle Dun surface's less coarse deposits (Doon fan gravels), which cover the most of the Dehradun area (Nakata 1972).

Chapter-3

METHODOLOGY

The work covers 2 sections 1.) Landuse/Landcover classification and 2) Accuracy Assessment. The LULC classification of the study area and accuracy assessment, were carried out as per the methodology presented below:-

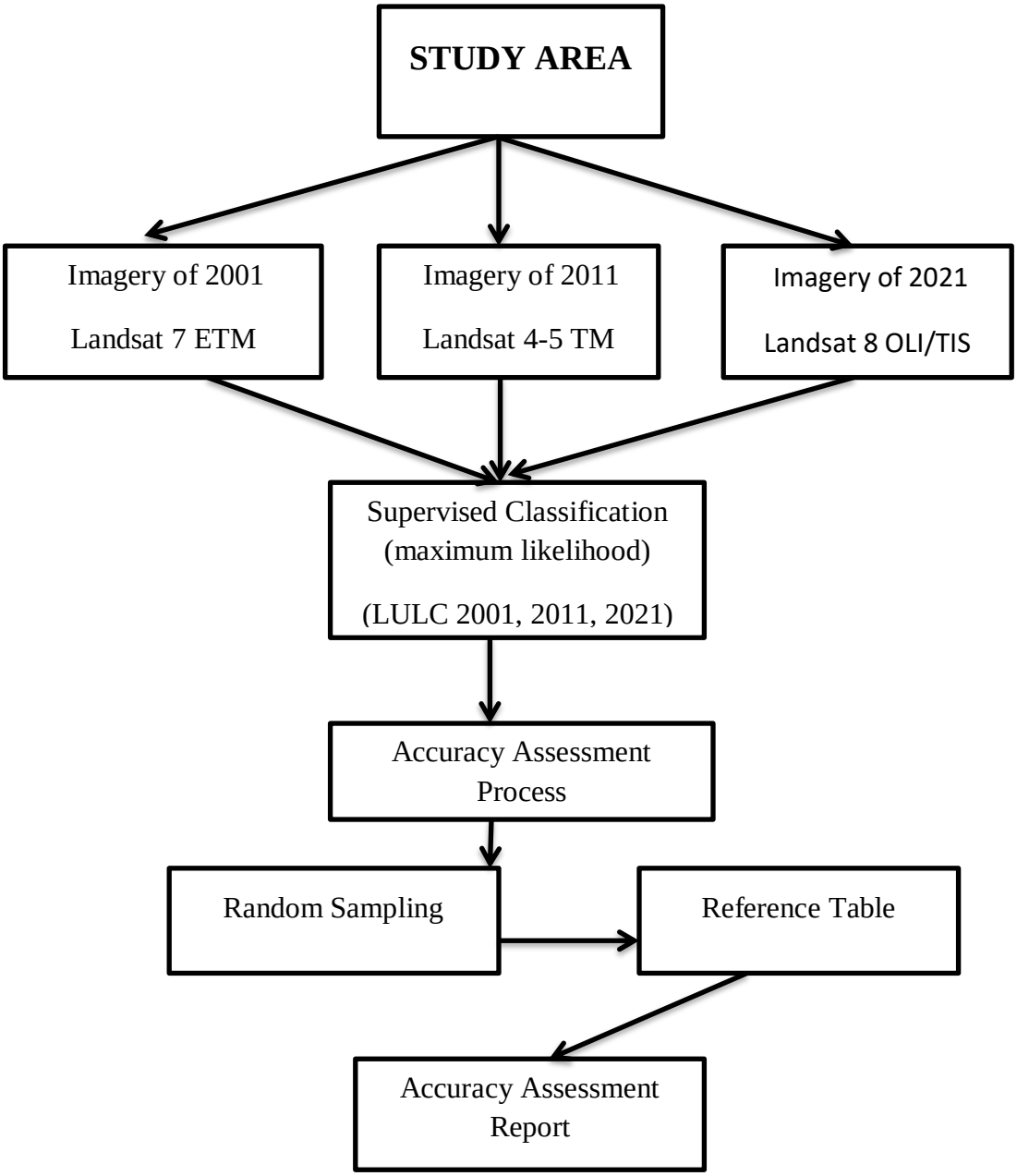


Fig 3 showing: Schematic of workflow of LULC followed by accuracy assessment

Classification process and analysis of the different LULC classes were done using the Landsat satellite images of the year 2001, 2011 and 2021. The data acquired for the year 2001 was from Landsat 7 ETM of 1st March 2001, for the year 2011 it was Landsat 4-5 TM of 25th February, 2011 and for the year 2021 it was Landsat-8 OLI/TIS (path 146, row 39) of 28th February, 2021. The Landsat imageries were downloaded form United States Geological (USGS) Earth Explorer. The images were selected on the basis of good quality image and minimum or no cloud cover. The Landsat images were georeferenced to the WGS_84 datum and Universal Transverse Mercator Zone (UTM) 44 North Coordinates System.

Satellite	**Sensor_ID**	**Path/row**	**Bands**	**Date of acquisiton**
Landsat 8 OLI/TIS	LC81460392021059LGN00	146/39	11	28th February, 2021
Landsat 5 TM	LT51460392011048KHC00	146/39	7	25th February, 2011
Landsat 7 ETM	LE71460392001060SGS00	146/39	9	1st March, 2001

Table 1: Details of Landsat data used for classification

Landuse/Landcover (LULC) Supervised Classification

In order to further proceed, the supervised classification was being carried out in ERDAS IMAGINE 2014 software. According to Eastman, J.R (2003) the supervised classification "the user develops the spectral signatures of known classes (for example, thick vegetation/forest, crop land, barren land, urban built-up) and then the software designates each pixels in the image to the class which resembles most to the provided signature". We first defined the training sites then extracted signature file after which it was applied to the composite Landsat images. The steps that were being followed are as follows:-

1.) Defining of training sites:

The first step of supervised classification is to identify and define the areas with ideal spectral characteristics of certain class that are further used as training sites. The Band layer-stacking was created as a composite raster file in the ArcGIS from the Landsat data

acquired (Table 1). Then, the study area of Dehradun was extracted from the 2001, 2011 and 2021 data.

2.) Extraction of Signatures –

The process was carried out in the ERDAS IMAGINE 2014, by defining 5 major classes- Thick vegetation, Cropland, Water body/ River, Barren land and Urban/Built-up.

Table 2: Landcover classification scheme

Land Cover	**Description**
Thick Vegetation	Land dominated by trees, mixed forest.
Crop Land	Lands covered with temporary crops followed by harvest period, crop fields and pastures
Barren Land	Land with exposed soil, sand, or rocks and never has more than 10% vegetated cover. Bare ground, bare rocks strip miners, quarries and gravel pits.
Built up areas	Land covered by buildings and other man-made structures. Residential, industrial area, mixed urban or built-up lands
Water Body	Lakes, Reservoirs, Stream, Rivers, Swamps

Further, approximately 50 polygons for each class were defined over the composite raster image of the study area. These 50 polygons in each class were then merged together under the name of their particular class.

After the entire signatures have been created, the signature file (.SIG) was saved for the use during supervised classification.

3.) Classification of the Image- using maximum likelihood classfier

The signature file was applied over the image of the study area and it gave the supervised classification image. The process of supervised classification was done using Maximum Likelihood Classifier for all the dataset.

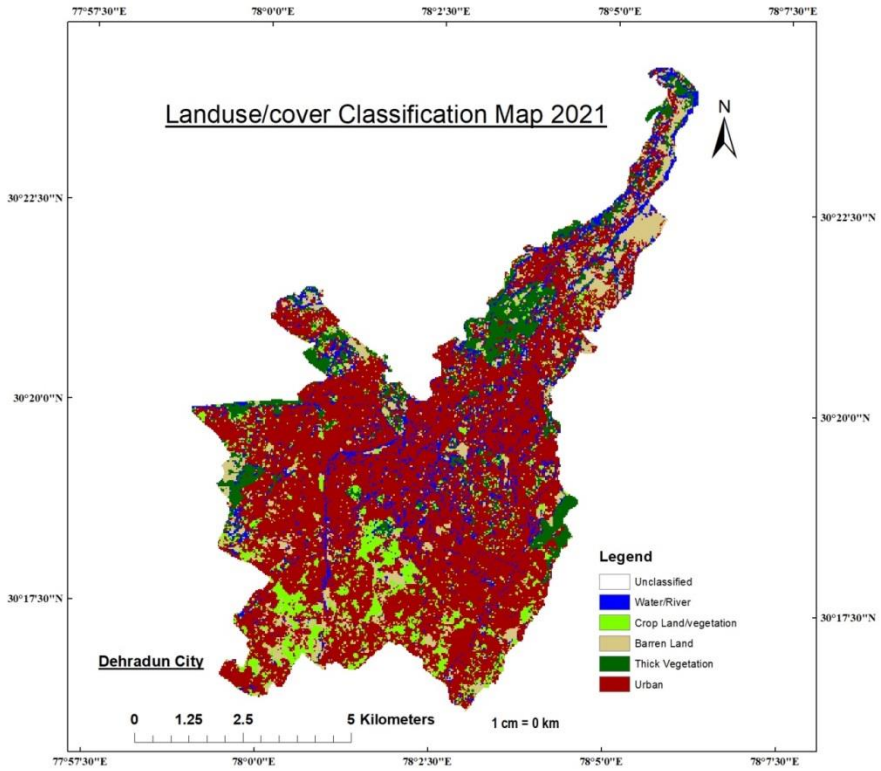


Fig 4- Classified map of the study area for the year 2021.

4.) Classification Accuracy Assessment Report and Error Calculation

The most important final step in the classification process is the accuracy assessment. The purpose of accuracy assessment is to assess how effectively the pixels were sampled into their correct land cover classes.

The process of random ground point selection was carried out in ArcGIS and were further divided into their respective classes manually, based on the user's knowledge. The random pixel selection taken on the Landsat data was also identified on the Google earth and the Landsat image.

Total of 75 ground points were defined for each map of the classified image (Fig. 5).



Figure 5: Ground Points on the study area

The attribute table data of random 75 points with their latitude-longitude coordinates and class-id in the table were saved as "reference file" into a ".txt" format. The supervised image was then opened in the ERDAS IMAGINE 2014. The accuracy assessment Cell Array Reference column was filled from the ".txt" file. And then the Accuracy Assessment report was generated which included error matrix, user's accuracy, producer's accuracy and overall accuracy.

Errors in classified map occurs because remotely-sensed data cannot capture classes that are not spectrally separable, atmospheric effects mask subtle differences between the pixels and spatial scale of remote sensing instrument does not match classification scheme.

The reason for the accuracy assessment to be low might be because of the 2 reasons: (1) Errors in reference data, (2) Errors in classified map (R. G. and K. Green. 1999).

Errors in reference data can be caused because of (a) Positional error- better rectification of the image may help reduce this error (b) Interpreter error (R. G. and K. Green. 1999).

Table 3: Error matrix of the LULC classification of the year 2011

Sr. No	Classified	Water Body	Thick Vegetation	Crop Land	Built-up	Barren Land	Total	Correct sampled	User's Accuracy
1	Water Body	6	0	0	3	1	10	6	60%
2	Thick Vegetation	0	15	0	0	0	15	15	100%
3	Crop land	1	0	15	0	1	17	15	88.2%
4	Built-up	3	0	0	15	0	18	15	83.3%
5	Barren land	0	0	0	0	15	15	15	100%
	Total	10	15	15	18	17	75	66	
Producer's Accuracy		60%	100%	100%	83.3%	88.2%			**Total=88%**

The table no. 3 shows the relationship between the ground truth data and corresponding classified data obtained through error matrix report.

The overall classification accuracy =

[No. of correct points / total no. of points] = (66/75)*100 = 88%

The table no. 3 is a theoretical error matrix of the LULC Classification of the year 2011. The rows of the error matrix show to which classes the image pixels have been assigned to in the image and the columns show to which classes the pixels is in with validation to those in the rows.

The pixels that are classified correctly according to reference data are shown diagonally. Pixels that are not assigned to the proper class or misclassified pixels occur off-diagonally and give an indication of the error between the different land-cover classes.

Furthermore, the off-diagonal elements in the rows of the error matrix, divided by the total number of pixels assigned to the Landsat image class corresponding to the row, represent the "commission error" and describes the confusion between that image class

and other land-cover classes. The commission errors describe the chance that a pixel that has been assigned to a particular class, actually belongs to one of the other classes.

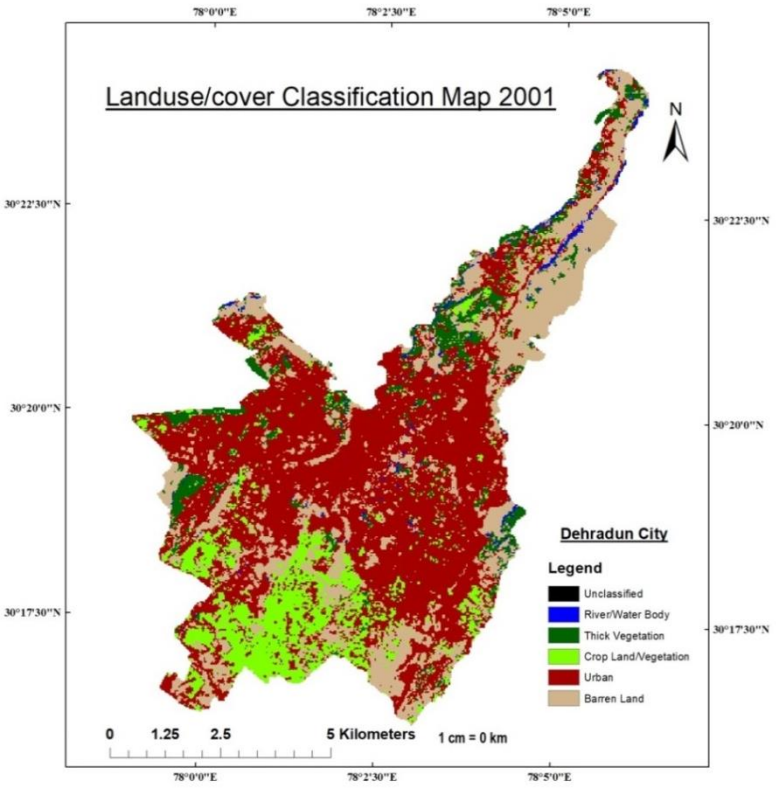
Choice of reference data is important in the process of accuracy assessment. Error matrix is the foundation of accuracy assessment. Interpreting accuracy in classes can yield ideas for improvement of the classification.

Chapter -4

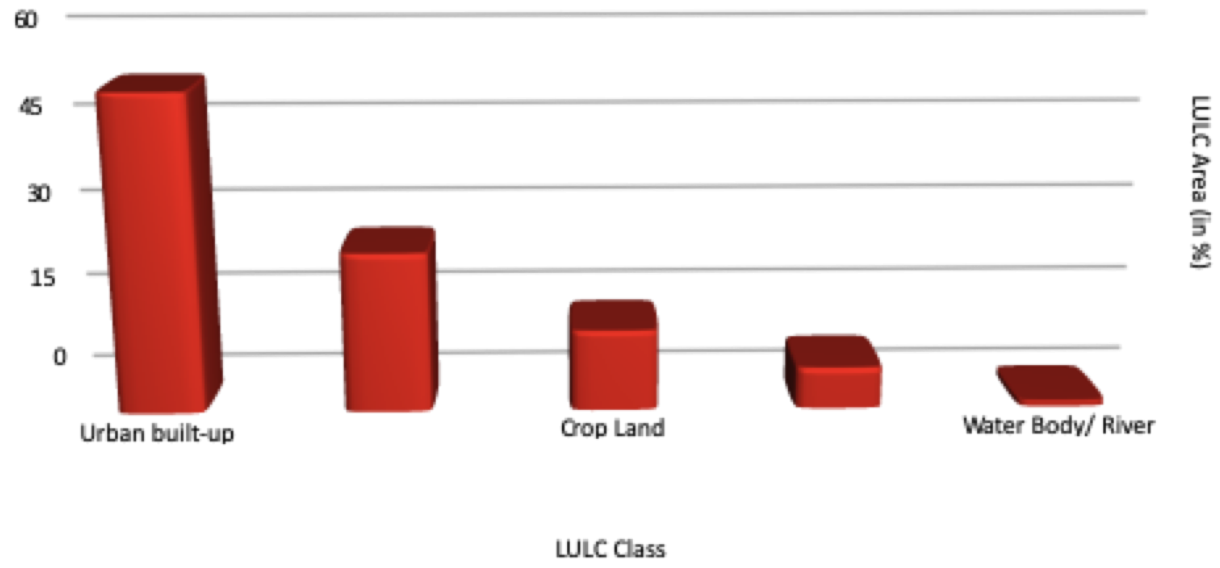
RESULTS AND DISCUSSION

LANDUSE AND LANDCOVER CLASSIFICATION FOR 2001 AND ACCURACY ASSESSMENT

The study area has been defined to have 5 major land use land cover categories, which were thick vegetation, cropland, barren land, water bodies, urban built-up. The description of these land cover categories was presented previously in Table 2. The land use land cover classification for 2001 showed that the majority of the study area was under urban built-up area 34.15 sq.km (52.31 %) followed by barren land, cropland, thick vegetation and water accounting 17.237 sq.km (26.40%), 8.75 sq.km (13.39%), 4.46 sq.km (6.83%) and 0.69 sq.km (1.05%) respectively.



(a)



(b)

Fig 6: (a) LULC map of Dehradun City-2001; (b) graph showing distribution of various classes.

Table 4- Land Use/Cover of 2001

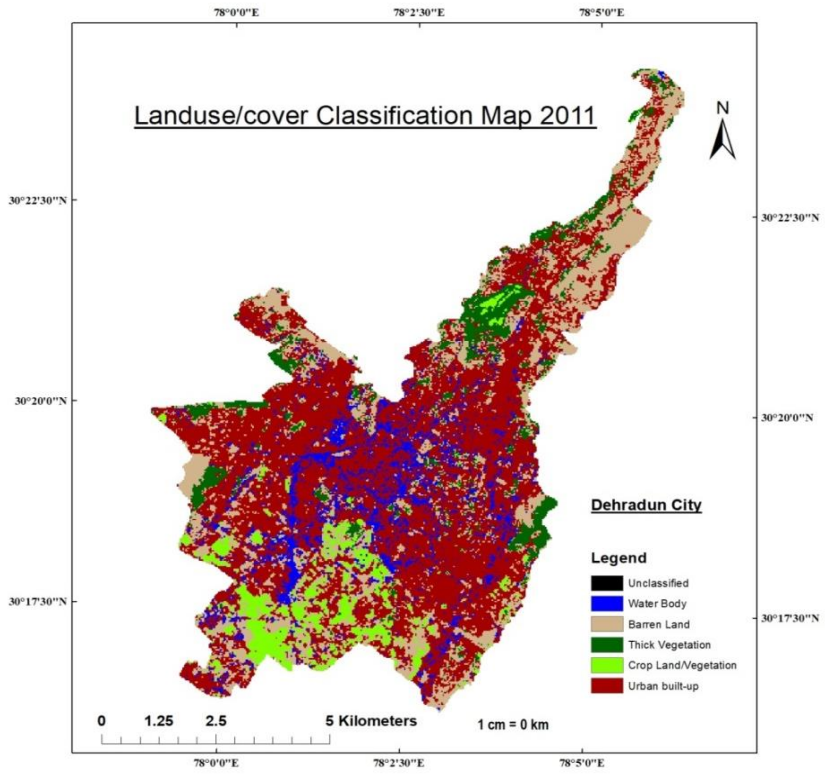
Land use/land cover Class	**Area in sq. km**	**Area in %**
Urban built-up	34.159	52.31
Barren land	17.237	26.40
Crop Land	8.744	13.39
Thick vegetation	4.461	6.83
Water Body/River	0.691	1.05
Total	65.29	100

Table 5 – Accuracy Assessment report of LULC 2001.

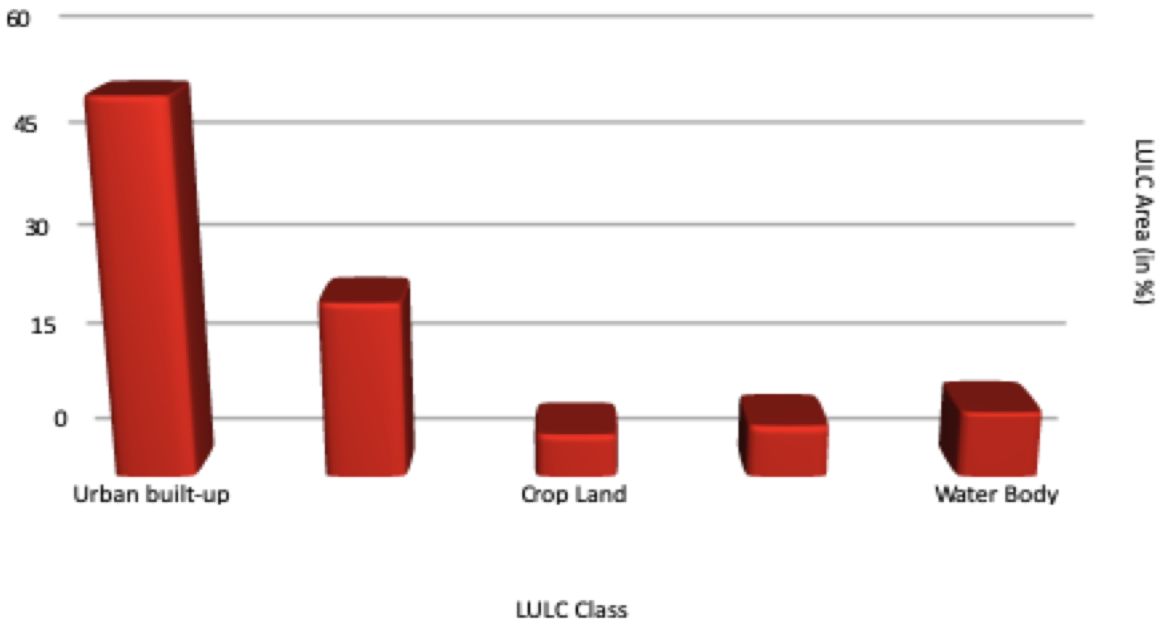
Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy	Kappa
Water Body	10	8	8	80%	100%	1.000
Thick Vegetation	15	14	13	86.67%	92.86%	0.9113
Crop Land	20	20	18	90.0%	90.0%	0.8649
Built-up	21	21	19	90.48%	90.48%	0.8690
Barren Land	10	13	10	100.0%	76.92%	0.7348
Totals	77	77	69			
Overall Classification Accuracy =No. of correct points/total number of points = **89.61 %** **Overall Kappa Statistics** = **0.868**						

LANDUSE AND LANDCOVER CLASSIFICATION FOR 2011 AND ACCURACY ASSESSMENT

The LULC classification for the 2011 shows that the area of urban built-up increases from 34.159 sq. km to 34.24 sq. km and consequently barren land decreases from 17.237 sq. km to 16.17 sq. km and crop land drastically decreases from 8.744 sq. km to 3.98 sq. km, whereas the thick vegetation observes a little increase from 4.461 sq. km to 4.81 sq. km from the year 2001.



(a)



(b)

Fig. 7: (a) LULC of Dehradun 2011; (b) graph showing distribution of various classes.

Land use/land cover Class	Area in sq. km	Area in %
Urban built-up	34.24	52.44
Barren land	16.17	24.77
Crop Land	3.98	6.09
Thick vegetation	4.81	7.36
Water Body/River	6.09	9.33
Total	65.29	100

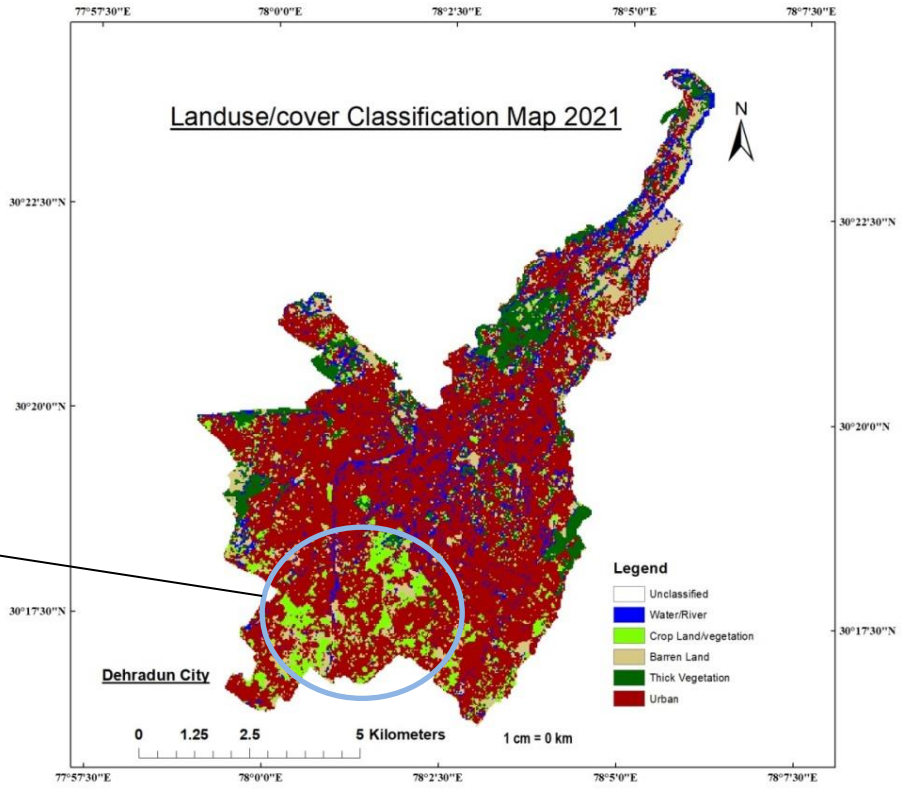
Table 6- Land Use/Cover data of 2011

Table 7 – Accuracy Assessment report of LULC 2011.

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy	Kappa
Water Body	10	10	6	60.0%	60.0%	0.5385
Thick Vegetation	15	15	15	100%	100%	1.000
Crop Land	17	15	15	88.24%	100.0%	1.000
Built-up	18	18	15	83.33%	83.33%	0.7807
Barren Land	15	17	15	100.0%	88.24%	0.8529
Totals	75	75	66			
Overall Classification Accuracy =(No. of correct points/total number of points)= **88.00 %** **Overall kappa Statistics** = **0.8489**						

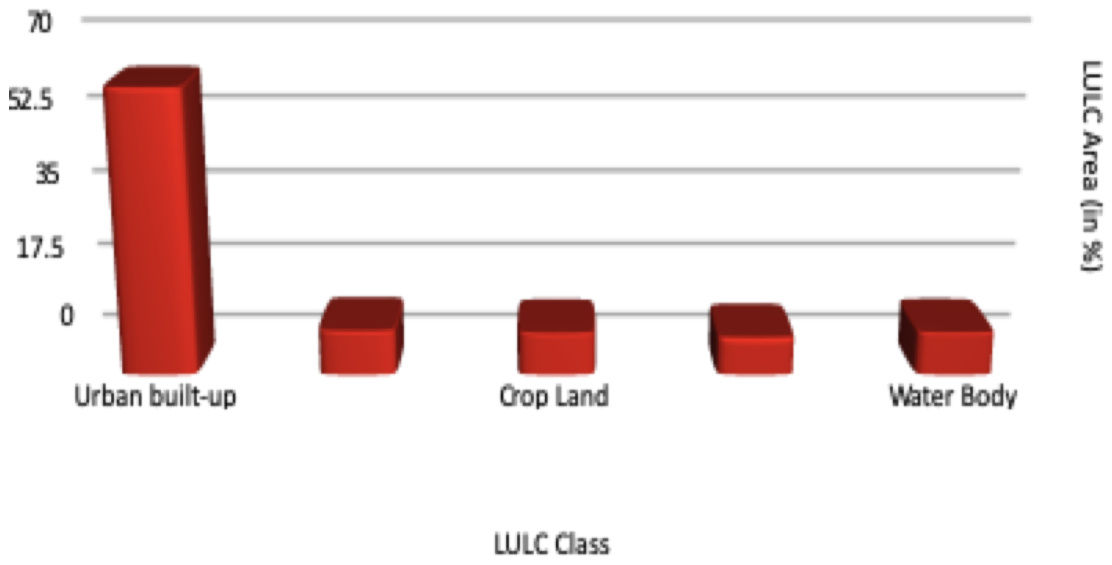
LANDUSE AND LANDCOVER CLASSIFICATION FOR 2021 AND ACCURACY ASSESSMENT

The LULC classification for the year 2021 shows that the area of urban built-up has further increased from 34.24 sq. km to 40.932 sq. km and barren land further decreases drastically from 16.17 sq. km to 6.485 sq. km, Crop land increases from 3.98 sq. km to 6.139 sq. km, Thick vegetation or forested area increases from 4.81 sq. km to 5.53 sq. km and water body also increases slightly from 6.092 sq. km to 6.209 sq. km from the year 2011.



Observed decrease in crop land and barren land due to the increase in the built-up area in the last 2 decades in this region.

(a)



(b)

Fig. 8: (a) LULC of Dehradun 2021; (b) graph showing distribution of LULC classes.

Table 8: Land Use/Land Cover 2021.

Land use/land cover Class	**Area in sq. km**	**Area in %**
Urban built-up	40.932	62.67
Barren land	6.485	9.93
Crop Land	6.139	9.4
Thick vegetation	5.53	8.4
Water Body/River	6.209	9.5
Total	65.29	100

Table 9 – Accuracy Assessment report of LULC 2021.

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy	Kappa
Water Body	10	11	10	100.00	90.91%	0.8951
Thick Vegetation	14	14	14	100%	100%	1.000
Crop Land	16	15	15	93.75%	100.0%	1.000
Built-up	20	20	19	95.00%	95.00%	0.9318
Barren Land	15	15	15	100.0%	100.00%	1.000
Totals	75	75	73			
Overall Classification Accuracy =(No. of correct points/total number of points)= **97.33 %** **Overall kappa Statistics** = **0.9663**						

The Table (10) and Graph below represents the overall comparison of area distribution among different classes in the year 2001, 2011 and 2021.

LULC Class	2001		2011		2021	
	Area (Sq. km)	Area %	Area (Sq. km)	Area %	Area (sq. km)	Area %
Urban Built-up	34.159	52.31	34.24	52.44	40.932	62.67
Barren Land	17.237	26.40	16.17	24.77	6.485	9.93
Crop land	8.744	13.39	3.98	6.09	6.139	9.4
Thick vegetation	4.461	6.83	4.81	7.36	5.53	8.4
Water Body	0.691	1.05	6.09	9.33	6.209	9.5
Total	65.29	100	65.29	100	65.29	100

Table- 10: Area covered by different classes in various years.

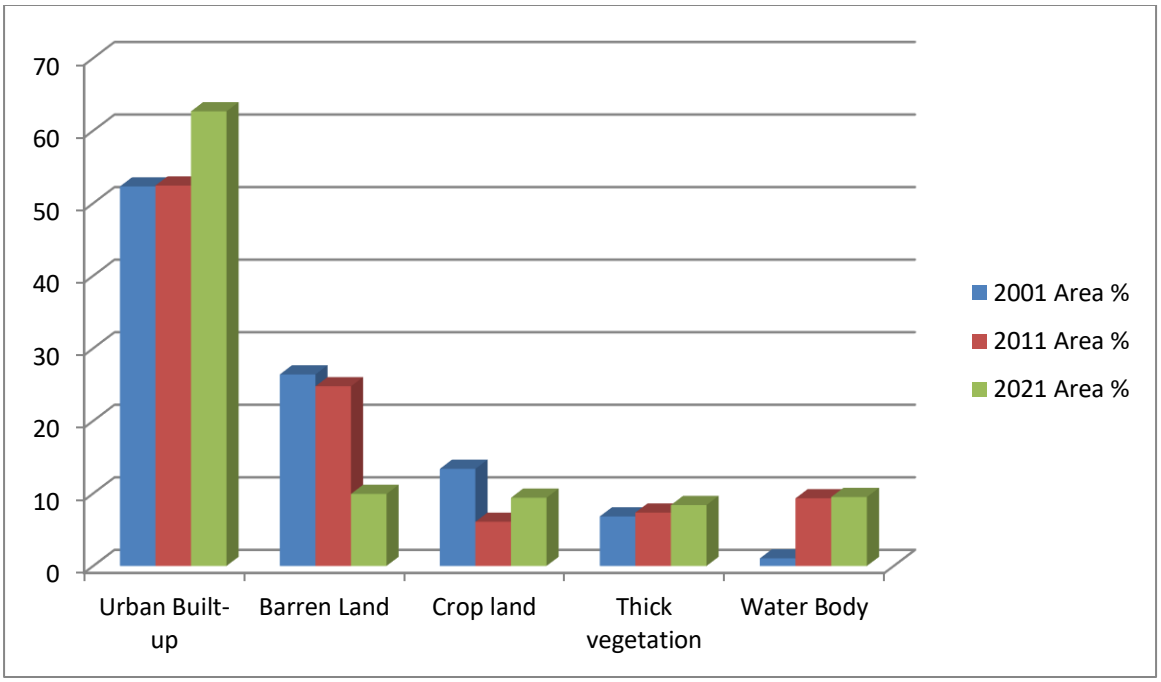


Figure 9: Graph showing comparison of area covered by various classes in three different years.

LULC Change 2001- 2011

Table 11:- Percentage change in different classes between 2001 and 2011.

LULC Class	**2001 Area in %**	**2011 Area in %**	**Change Area %**
Urban built-up	52.31	52.44	+0.13
Barren land	26.40	24.77	-1.63
Crop Land	13.39	6.09	-7.3
Thick vegetation	6.83	7.36	+0.53
Water Body	1.05	9.33	+8.28
Total	100	100	

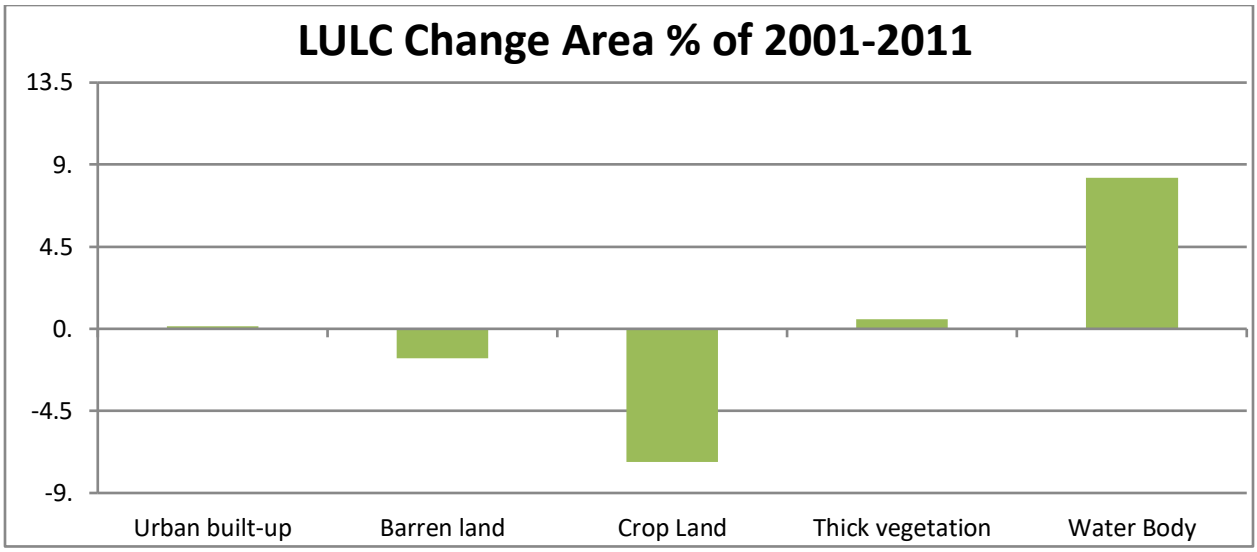


Fig 10: Graph showing percentage change of area by different classes between 2001 and 2011.

LULC Change 2011-2021

Table 12:- Percentage change in different classes between 2011 and 2021.

LULC Class	**2011 Area in %**	**2021 Area in %**	**Change Area %**
Urban built-up	52.44	62.67	+10.23
Barren land	24.77	9.93	-14.84
Crop Land	6.09	9.4	+3.31
Thick vegetation	7.36	8.4	+1.04
Water Body	9.33	9.5	+0.17
Total	100	100	

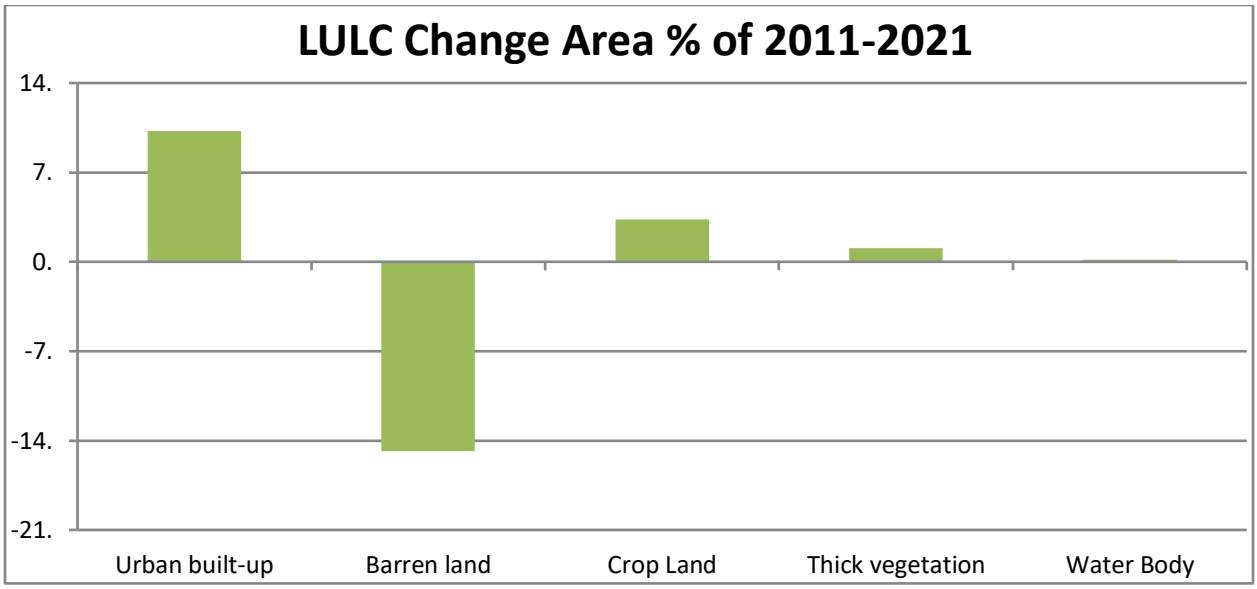


Fig 11: Graph showing percentage change of area by different classes between 2011 and 2021.

LULC Change 2001 – 2021

Table 13:- Percentage change in different classes between 2011 and 2021.

LULC Class	**2001 Area in %**	**2021 Area in %**	**Change Area %**
Urban built-up	52.31	62.67	+10.36
Barren land	26.40	9.93	-16.47
Crop Land	13.39	9.4	-3.99
Thick vegetation	6.83	8.4	+1.57
Water Body	1.05	9.5	+8.45
Total	100	100	

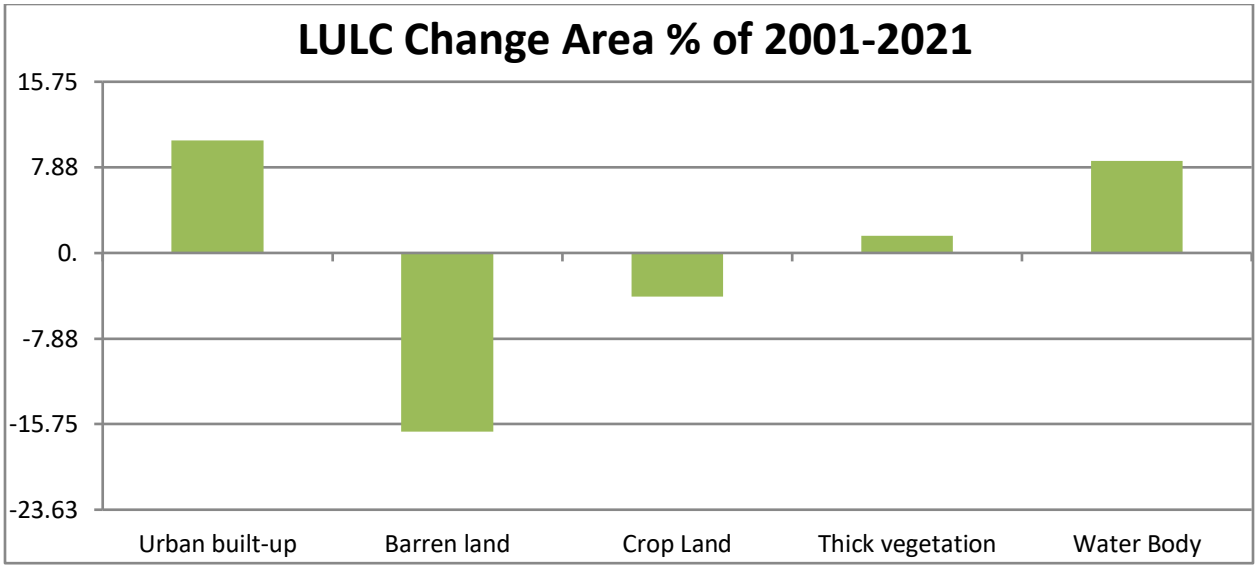


Fig 12: Graph showing percentage change of area by different classes between 2001 and 2021.

The proportion of total population in a region to the total built-up is a measure of quantifying city growth (Rahman et al., 2011). In the year 2000 Dehradun was declared capital city of the newly formed state of Uttarakhand. The *increase* of built-up area during the period 2001-2021 was about 10.36 %. The increasing population of the city needs areas to construct the houses for their living and therefore changes in land use pattern becomes inevitable. It is evident that the city built-up has expanded towards its periphery to cater the demand of growing population in the city. The area of vegetation and barren land have observed overall major *decrease* of 3.99% and 16.47% respectively in the last 2 decades. This depicts the process of conversion from vegetated land to urban land which again indicates the growth of city further towards its outskirts (Fig. 8a). This is mainly due to increasing job opportunities in the state capital which raised the inflow of migrants, making the city congested within a very short span of time. Table no. 13 clearly supports the above statements. However, in the area of thick vegetation a gradual increase from 6.83% to 7.36% from 2001 to 2011 and up to 8.4% in the year 2021 has been observed.

The rapid process of urbanization is the main cause for the conversion of Doon valley floor into an overloaded residential area with shrinking crop land and barren land (little or no vegetated land) and problem regarding environment degradation in the valley.

Chapter-5

CONCLUSION

In the year 2000 Uttarakhand became a state of the Republic of India after separating from Uttar Pradesh, with Dehradun as its capital city. Since then, Dehradun has been under rapid advancement. The city's greatest land infringement activities for the developmental purpose like infrastructure, roadways, housing and utility has taken place after the year 2000. The city has experienced rapid growth of population because of increasing rural to urban migration during the last 2 decades. This in turn escalated the demand for land for various urban activities in the city. Therefore, a rapid decrease in the area of barren land and crop land has occurred within the last 2 decades to accommodate increasing number of houses. The infringement of agricultural land, barren land and forest land has taken place for infrastructure advancement. The outskirts of the city has observed little or no left barren land due to increasing population and rapid urbanization.

REFERENCES

Sophia S. Rwanga, J.M. Ndambuki (2017). Accuracy Assessment of Land Use/Land Cover Classification Using Remote Sensing and GIS, International Journal of Geosciences, 8 611-622.

A. K. Mahajan, Siefko Slob & Rajiv Ranjan & Rob Sporry & P. K. Champati ray & Cees J. van Westen (2007). Seismic microzonation of Dehradun City using geophysical and geotechnical characteristics in the upper 30 m of soil column, J Seismol (2007) 11:355–370 DOI 10.1007/s10950-007-9055-1.

Maya Kumari (2014). Change detection analysis using multi temporal satellite data of Poba reserve forest, Assam and Arunachal Pradesh, International Journal of Geomatics and Gesciences Volume 4, No 3, 2014, ISSN 0976-4380.

Prakash P (2018). Study the 2013 Flood damages and Risk Assessment in Kedarnath, Himalya Area using Geoinformatic Techniques, Journal of Geography & Natural Disasters, DOI: 10.4172/2167-0587.1000216

Swati Sinha, Rajiv Sinha (2016). Geomorphic evolution of Dehra Dun, NW Himalaya: Tectonics and climatic coupling, Department of Earth Sciences, Indian Institute of Technology, Kanpur 208016, India.

Sunil Sankhala, B.K. Singh (2014). Evaluation of Urban Sprawl and Land use Land cover Change using Remote Sensing and GIS Techniques: A Case Study of Jaipur City, India, international Journal of Emerging Technology and Advanced Engineering. ISSN 2250-2459, ISO 9001:2008 Certifies Journal.

Dipanwtta K Dutta, Arnab KUNDU, Atqir Rahman (2015). Growth of Dehradun City: An application of linear spectral unmixing (LSU) technique using multi-temporal landsat satellite data sets, Vidyasagar university, Mindapore, India

Kshama Gupta(2013), Unprecedented Growth of Dehradun Urban Area: A spatio-temporal analysis, Indian Institute of Remote Sensing, Dehradun, India, International Journal of Advancement in Remote Sensing, GIS and Geography, ISSN 2321-8355

Eastman, J.R (2003) Guide to GIS and Image Processing 14, 239-247. Clark University Mannual, USA

Congalton, R.G. (1991) A Review of Assessing the Accuracy of Classifications of Remotely Sensed Data. 0034-4257(91)90048-B

Banerjee, A., Reddy, K.N., Paul, P., 2002. Application of remote sensing and GIS in demographic and socio-economic analysis of Dehradun city. Indian Cartogr.22, 274–281.

Jain, S., Jain, R.K., 2006. A remote sensing approach to establish relationships among different landcovers at the micro level. Int. J. Remote Sensing 27, 2667–2682.

Rahman, A., Kumar, Sunil, Fazal, Shahab, Siddiqui, Masood A., 2012. Assessing landuse/Landcover change in the North-West District of Delhi using remote sensing and GIS techniques. J. Indian Soc. Remote Sens. 40(4). (ISSN:No.0255-660).

Rahman, A., Aggarwal, S.P., Netzband, M., Fazal, S., 2011. Monitoring Urban Sprawl Using Remote Sensing and GIS Techniques of a Fast Growing Urban Centre, India. IEEEJ. Sel. Top. Appl. Earth Obs. Remote Sens. (JSTARS) 4,56–64.

Nakata, T (1972) Geomorphic history and crustal movement of the foot-hills of the Himalayas, Sci. Report, Tohoku University, Japan, 7th Ser. Geography 22:39–177

Nossin, JJ (1971) Outline of geomorphology of the Doon valley, northern UP. Z Geomorphology 12:18–50

Thakur VC (1995) Geology of Dun Valley, Garhwal Himalaya: neotectonics and coeval deposition with fault—propagation folds. J Himal Geol 6(2):1–8

Raiverman, V., Kunte, S.V., Mukherjee, A., 1983. Basin geometry, Cenozoic sedimentation and hydrocarbon prospects in north-western Himalaya and Indo-Gangetic plains. Petrol. Asia J. 6,67-92

Raiverman, V., Srivastava, A.K., Prasad, D.N., 1994. Structural style in northwestern Himalayan foothills. Himal. Geol. 15, 263-280

Singh, A.K., 1998. Geomorphology, Sedimentology and Pedology of the Intermontane Dehradun valley, Northwest Himalaya (Unpublished PhD. Thesis) University of Roorkee

Singh, A.K., Prakash, B., Mohindra, R., Thomas, J.V., Singhavi, A.K., 2001. Quaternary alluvial fan sedimentation in the Dehradun Valley Piggyback Basin, NW Himalaya: tectonic and palaeoclimatic implications. Basin Res. 13, 449-471.

Congalton, R. G. and K. Green. 1999. Assessing the accuracy of remotely sensed data: Principles and practices. Lewis Publishers, Boca Raton.